

Lecture 23

- ↳ Grover Search
- ↳ Modify for Multiple Entries.
- ↳ Use of Grover for NP Problems.
- ↳ Study of Quantum Error Correction.

Suppose you find $|\vec{y}_3\rangle$,

You have state $|\Phi\rangle$

measure $|\vec{y}_3\rangle\langle\vec{y}_3|$

↳ if you get

$$|\Phi\rangle = \sqrt{c}|\vec{y}_3\rangle + \sqrt{1-c}|\chi\rangle$$

get a state with $|\vec{y}_3\rangle$ removed

$|\vec{y}_3\rangle\langle\vec{y}_3||\Phi\rangle =$
 ↳ if you measure & get 0,
 if you measure 1 → unfortunate.

this is not unitary

$$U_f = I - 2|\vec{y}\rangle\langle\vec{y}|$$

↳ looks

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - 2 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = "Z"$$

iZ

is a rotation by π around the Z -axis.

Z is a reflection.

$$P = I - |\psi\rangle\langle\psi|$$

$$P|\psi\rangle = -|\psi\rangle$$

removes $|\psi\rangle$ from the superposition

$$(I - \hat{n} \cdot \vec{\sigma})$$

(can you remove a state from a superposition)

BBBV Theorem: $(\sqrt{N})$ speedup is optimal

Extension to Multiple Entry Search.

$$f(x) = 1 \text{ if } x \in \{\vec{y}_1, \vec{y}_2, \dots, \vec{y}_m\}$$

instead of a target $|\vec{y}\rangle$

let's seek

$$|T_m\rangle = \frac{1}{\sqrt{M}} \sum_{\vec{x}} f(\vec{x}) |\vec{x}\rangle$$

$$U_f = I - 2 \sum_{j=1}^M |\vec{y}_j\rangle \langle \vec{y}_j|$$

$$U_f \neq V_f \equiv I - 2|T_M\rangle \langle T_M|$$

(?) there are nondisjoint terms

Nevertheless

$$U_f |T_M\rangle = V_f |T_M\rangle = -|T_M\rangle$$

$$U_f \left(\frac{1}{\sqrt{M}} \sum_{j=1}^M |\vec{y}_j\rangle \right) = -|T_M\rangle = U_f |T_M\rangle$$

$$\Rightarrow |\Phi_0\rangle = \sqrt{M} c |T_M\rangle + W_M |X_M\rangle$$

$$\hookrightarrow \text{where } |X_M\rangle \perp |T_M\rangle$$

$$\langle T_M | \Phi_0 \rangle = \sqrt{\frac{M}{2^n}}$$

$$= \sqrt{M} c$$

$$U_f |\Phi_0\rangle = V_f |\Phi_0\rangle$$

$$N_{\text{queries}} \approx \frac{\pi}{4} \sqrt{\frac{2^n}{M}}$$

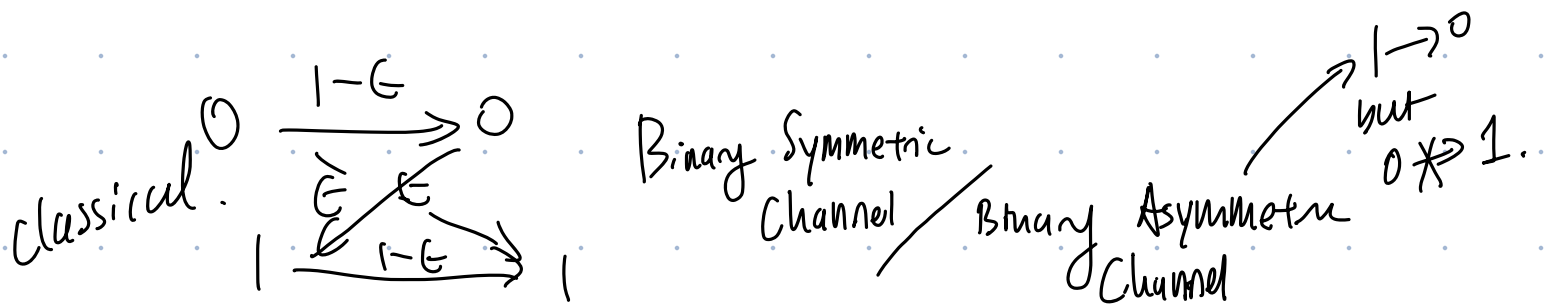
Oblivious Amplification

↳ Does not overshoot

(once you get to the south pole, you don't move around that much).

→ PRX

Quantum Error Correction:



quantum

X

$$Z \quad |0\rangle + |1\rangle \rightarrow |0\rangle - |1\rangle$$

$$e^{i\frac{\theta}{2}Z} \quad |0\rangle + |1\rangle \rightarrow \cos\frac{\theta}{2} (|0\rangle + |1\rangle) - i\sin\frac{\theta}{2} (|0\rangle - |1\rangle)$$

$\nearrow$

$$\cancel{\theta} \quad Y = iXZ$$

$\exists$ a code to detect errors.